Automated Quality Assurance of Gamified Tutorials in a Quality Management Web Application Using LLM-Driven End-to-End Testing and Mutation Testing

Master's Thesis Exposé

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Abstract

Modern quality management web applications evolve continuously and require reliable automated testing to ensure the correctness of critical workflows. In industrial environments, end-to-end tests are commonly used to validate user-facing processes, yet maintaining their quality over time remains challenging due to frequent changes in interfaces and logic. This thesis investigates an automated quality assurance approach for a cloud-deployed Quality Management web application at Infineon, which includes gamified tutorial elements that guide users through complex workflows during onboarding and first use.

The focus of the work is on assessing and improving the quality of end-to-end tests for such UI-driven workflows. The proposed approach combines end-to-end testing using Playwright with mutation testing using Stryker to evaluate test adequacy. To support test creation and maintenance, an LLM-based test generation and refinement process is introduced, instantiated using GitHub Copilot. Test execution results and mutation testing feedback are used as structured input to iteratively improve generated test cases.

The thesis follows a Design Science Research approach. The developed testing process is evaluated empirically by comparing LLM-assisted test suites with manually written and recorded tests using metrics such as mutation score, failure detection capability, and maintenance effort. The evaluation is conducted on a real industrial system to ensure practical relevance.

The results contribute to research on AI-assisted software testing by examining whether a feedback-driven LLM approach can systematically improve the quality of end-to-end tests in web-based quality management systems.

Introduction

Game Studies and Engineering examines how digital systems and design decisions influence user behavior, learning, and interaction. One recurring topic in this field is how interactive and game-inspired approaches can support users when they must understand complex systems or unfamiliar workflows. While much existing research focuses on educational or entertainment contexts, similar challenges arise in professional environments where employees rely on specialized digital tools.

In technical workplaces, internal software tools evolve continuously and must remain reliable despite frequent changes. At the same time, these systems are often supported only by informal explanations or limited documentation. This combination creates two related challenges: users struggle to learn and

correctly apply complex workflows, and developers must ensure that user-facing processes remain functionally correct over time. As a result, quality assurance and automated testing become critical, particularly for web-based systems that integrate interactive guidance directly into operational workflows.

The thesis focuses on an internal chip-grouping web application that includes gamified tutorial system to support users during task execution. The core contribution of the thesis lies in the design and empirical evaluation of automated testing approaches that ensure the correctness and robustness of these guided workflows as the system evolves.

Motivation

The selection of this research topic comes from direct experience gained during my involvement in developing a semiconductor chip grouping tool for Infineon's Quality Management department. The tool was designed to support employees in daily operations by providing a unified solution for a workflow that is inherently complex and varies depending on user context and available information. Prior to the tool's development, this workflow required the use of multiple platforms and manual coordination. The chip grouping approach aims to reduce the number of required *test vehicles*[1] through efficient device grouping strategies, thereby accelerating monitoring, testing, and downstream processes.

The workflow handled by the tool is complex and requires several decisions and data inputs, because of that, it was already clear during the design phase that users would need guidance while using the system. The process cannot be described as a single fixed sequence of steps and may differ depending on the situation. For this reason, simple written documentation was not considered sufficient. Instead, guided tutorials directly integrated into the application were planned as a practical way to support users during task execution.

At the same time, embedding interactive tutorials directly into operational workflows introduces new challenges from a software engineering perspective. As the system evolves, both the workflows and the associated guidance must remain correct and consistent. Ensuring this requires reliable quality assurance mechanisms, particularly automated testing approaches that can validate user-facing processes over time.

From a research perspective, this situation reflects a broader challenge in technical departments: complex internal workflows are increasingly consolidated into web-based tools, while systematic methods to test and maintain interactive guidance remain limited. This thesis addresses this gap by

[1] A test vehicle is a selected semiconductor device used to evaluate reliability of materials, manufacturing processes, and design characteristics, often based on standards such as JEDEC

focusing on the design and evaluation of automated testing strategies for guided workflows in an industrial software system.

Goal and Research Questions

The overall goal of this thesis is to design and evaluate a quality assurance approach for a gamified, tutorial-based workflow implemented in a quality management web application at Infineon. The application supports a complex chip-grouping process and integrates tutorials that guide users step by step through the workflow. Since these tutorials are a core part of the application, their correctness, stability, and long-term quality must be ensured as the system evolves. The thesis therefore focuses on how automated testing and feedback-driven improvements can be used to maintain and improve both the application and its tutorial workflows over time.

To achieve this goal, the thesis investigates automated end-to-end testing for validating tutorial-driven workflows and examines how feedback from testing and usage can be used to improve the system. An LLM is used as a supporting component to help analyze feedback and guide improvements to the application and tutorials, rather than to generate tests directly. In addition to technical testing results, user feedback is collected to assess the overall effectiveness of the tutorial system in practice.

Based on this goal, the thesis addresses the following research questions:

1. **How should automated end-to-end testing be designed to check the correctness of gamified tutorial workflows in a quality management web application?**

2. **Can feedback-based improvements supported by an LLM help improve the quality of the application and tutorials over time?**

3. **Can this approach also lead to better tool quality and user engagement, based on testing results and user feedback?**

Literature Review

Automated testing is the main method for ensuring the quality of modern web applications, especially in industrial systems that evolve continuously. In such systems, end-to-end testing is commonly used to validate complete user workflows from the user’s perspective. Recent surveys show that while end-to-end tests are effective at detecting integration and workflow errors, they are also costly to maintain due to frequent changes in user interfaces and application logic (Kertusha et al. 2025; Maunu 2025).

Research in software testing highlights the need for better methods to assess and improve test quality beyond basic coverage metrics. Studies on automated testing in continuous integration environments show that systematic test evaluation helps detect errors earlier and improves system reliability over time (Afrihyia 2025). In practice, this requires test suites that remain stable and meaningful as systems change, particularly for user-facing workflows.

Gamification has been studied as a way to support users in complex task environments. Gamification is commonly defined as the use of game design elements in non-game contexts (Deterding et al. 2011). Empirical research shows that gamified approaches can also be applied in industrial and manufacturing contexts when they are embedded into real workflows rather than used only for training (Hamari, Koivisto, and Sarsa 2014; Capponi et al. 2024). However, most studies focus on learning or engagement outcomes, and little attention is given to how gamified guidance integrated into operational systems can be tested and maintained.

Recent research also explores the role of large language models in software engineering. Surveys describe LLMs as support tools that can help analyze software artifacts such as test results and logs, rather than replacing existing testing techniques (Zhao et al. 2023). This aligns with approaches that use LLMs to support feedback-driven improvement of software quality.

Overall, existing research provides strong foundations in automated web testing, end-to-end workflow validation, and gamification. However, there is limited work that combines these areas to study quality assurance of gamified, tutorial-based workflows in industrial web applications. This thesis addresses this gap by evaluating an automated testing-centered approach supported by structured feedback in a real quality management system.

Research Methodology

This thesis follows a Design Science Research (DSR) approach, which is suitable for designing and evaluating technical solutions in real-world software systems (Hevner et al. 2004). DSR provides a structured framework for developing an artifact and assessing its effectiveness in practice..

Evaluation is based on automated testing data and user feedback. End-to-end tests are used to validate complete user workflows, and their results are analyzed using quantitative metrics such as test pass rates, recurring failures, and execution stability. Mutation testing is applied as a complementary technique to assess test adequacy, using mutation scores as an indicator of how well the tests detect relevant errors.

User feedback is collected to evaluate the quality and clarity of the guided tutorials. This includes measures such as perceived usefulness, ease of understanding, and time spent completing tutorial-guided workflows. These results are used to compare tutorial effectiveness across iterations and to complement the findings from automated testing.

Testing results, mutation feedback, and user feedback are analyzed together to identify weaknesses in both the application and the tutorials and to guide subsequent design improvements.

Prototype Concept

The gamification concept in this project focuses on structured, embedded guidance rather than explicit game mechanics. Based on management input, the priority is to provide an integrated tutorial system that supports employees during complex chip-grouping workflows. Elements such as points, levels, or badges are not included in the current scope and may be considered only in later iterations after evaluation.

The tutorials are integrated directly into the web-based chip-grouping tools of the Quality Management hub and guide users step by step within the application interface. The system is designed so that tutorial steps and workflow actions are directly observable and testable. This allows automated end-to-end tests to validate not only functional behavior, but also the correct execution of guided workflows.

The frontend is implemented using React, HTML, CSS, and Refine, while backend components include C#, Node.js, MySQL, and Denodo. Deployment and authorization are handled via HICP infrastructure and Amazon Redshift. Playwright is used to automate end-to-end tests, including tutorial-driven user interactions. Tutorial overlays are implemented using Shepherd.js or comparable libraries. A technical overview and example screenshots are provided in Appendix A.

The main benefit of this approach lies in the clear structuring of information during task execution, following a progressive disclosure principle. Only the information required for the current step is shown, reducing cognitive load and supporting learning through actual task performance.

From a quality assurance perspective, embedding tutorials directly into the workflow enables systematic testing of user guidance. Guided steps become part of the executable process and can be validated automatically as the system evolves, reducing the risk of broken or misleading tutorials over time.

Timeline

The thesis is planned for a period of six months, starting in **October 2025** and ending in **March 2026**. The schedule follows the Design Science Research cycle, which includes problem identification, design, development, testing, evaluation, and reporting.

October – Identify the research problem, complete literature review, and define the conceptual framework.

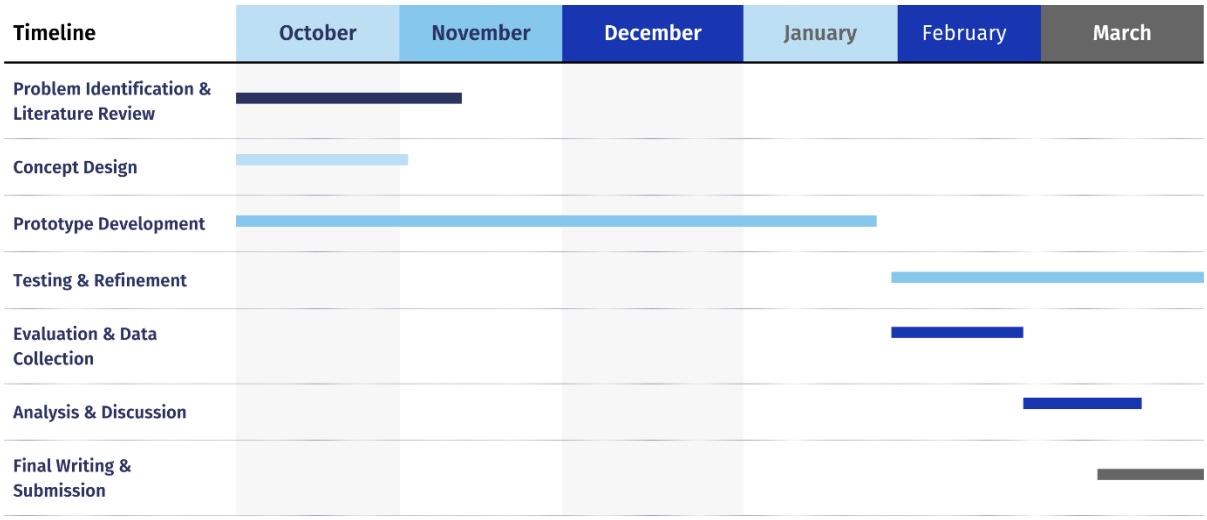
November – Prepare system design and confirm requirements through discussions with Infineon employees.

December to January – Continue development of the prototype. The first version of the tutorial system has already been built, and current work focuses on the backend for deployment, where data is fetched from a Denodo view, processed, and transferred into the required tool.

February – Conduct the main evaluation and collect both quantitative and qualitative data from users.

March – Analyze results, finalize the thesis document, and present outcomes.

The Gantt chart below visualizes the project timeline.



Bibliography

Anwar, Nahid, and Susmita Kar. 2019. “Review Paper on Various Software Testing Techniques & Strategies.” Global Journal of Computer Science and Technology: C Software & Data Engineering 19 (2).
https://www.researchgate.net/publication/336257973_Review_Paper_on_Various_Software_Testing_Techniques_Strategies

Capponi, Matteo, Angelo Nicola Culotta, Riccardo Gervasi, Luca Mastrogiacomo, and Fiorenzo Franceschini. 2024.“Enhancing Manufacturing Quality Through Gamification: An Exploratory Study in Collaborative Assembly Process.”
In *Proceedings of the International Conference on Quality Engineering and Management*, 861–888.
https://www.qualityengineering.polito.it/content/download/1171/6379/file/ICQEM_paper_final_gamification.pdf

Deterding, Sebastian, Dan Dixon, Rilla Khaled, and Lennart Nacke. 2011.
“From Game Design Elements to Gamefulness: Defining ‘Gamification.’”
In *Proceedings of the 15th International Academic MindTrek Conference*, 9–15.
https://doi.org/10.1145/2181037.2181040

Hamari, Juho, Jonna Koivisto, and Harri Sarsa. 2014.
“Does Gamification Work? A Literature Review of Empirical Studies on Gamification.”
In *Proceedings of the 47th Hawaii International Conference on System Sciences*, 3025–3034.
https://doi.org/10.1109/HICSS.2014.377

Kertusha, Iva, Gebremariem Assres, Onur Duman, and Andrea Arcuri. 2025.
“A Survey on Web Testing: On the Rise of AI and Applications in Industry.”
https://arxiv.org/abs/2503.05378

Maunu, Timo. 2025.
Optimizing the Creation and Maintenance of Automated End-to-End Tests in Agile Development.
Master’s thesis, Aalto University.
https://aaltodoc.aalto.fi/bitstreams/ad8ffe57-fc83-4ba3-982a-b242d7200655/download

Panwar, Arti, and Prasadu Peddi. 2023.
"Implementation of Software Testing Using Machine Learning: A Systematic Mapping Study."
SSRN Electronic Journal.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4899829

Afrihyia, Emanuel. 2025.
"Enhancing Software Reliability Through Automated Testing in CI/CD."
Frontiers in Multidisciplinary Research 2: 029.
https://www.multidisciplinaryfrontiers.com/uploads/archives/20250721163044_FMR-2025-2-029.1.pdf

Zhao, Wayne Xin, Kun Zhou, Junyi Li, Tian Tang, Xiaoyang Wang, Yupeng Hou, Yuxiao Min, Beichen Zhang, Junjie Zhang, and Zhendong Dong. 2023.
"A Survey of Large Language Models."
https://arxiv.org/abs/2303.18223

Appendix A – Tech Stack Diagrams

Figure A.1: Technical Architecture of the Quality Management Hub and Guided Tutorial System



Figure A.2: Example of Guided Tutorial Flow Using Shepherd.js

